

Displays

D1:

$$\Pr[y_{m+1} \in C(x_{m+1})] \geq 1 - \alpha.$$

D2:

$$\Pr \left[y_{m+1} \in [\mu(x_{m+1}) - R^*, \mu(x_{m+1}) + R^*] \right] \geq 1 - \alpha.$$

D3:

$$\Pr [\sigma_{s_0} \in X] \geq 1 - \varepsilon \quad (1)$$

D4:

$$\Pr[\sigma_{s_0} \in X] \geq \Delta \quad (2)$$

D5:

$$\bar{\sigma}_{s_0} = \mathcal{F}(s_0) = \left[s_0^\top, \mathbf{F}^1(s_0), \dots, \mathbf{F}^n(s_0), \dots, \mathbf{F}^{(n-1)K}(s_0), \dots, \mathbf{F}^{nK}(s_0) \right]^\top$$

D6:

$$\mathcal{I}_i \subset \mathcal{I}, \quad \bigcup_{i=1}^N \mathcal{I}_i = \mathcal{I},$$

D7:

$$R^j = |e_{j+n}^\top \sigma_{s_0} - \mathbf{F}^j(s_0)|$$

D8:

$$\mathcal{R}_{D_{\text{test}}}^j = \left\{ \left(s_{0,i}, R_i^j \right) \mid (s_{0,i}, \sigma_{s_{0,i}}) \in D_{\text{test}}, i \in [L] \right\}.$$

D9:

$$\Pr[R^j \leq R^{j*}] \geq \delta.$$

D10:

$$X = \bar{X} \oplus \text{Zonotope}(0, \mathbf{diag}([0_{1 \times n}, R^*])), \quad R^* = [R^{1*}, \dots, R^{nK*}].$$

D11:

$$\Pr[R^j > R^{j*}] < 1 - \delta.$$

D12:

$$\Pr \left[\bigvee_{j=1}^{nK} (R^j > R^{j*}) \right] < nK(1 - \delta).$$

D13:

$$\Pr \left[\bigwedge_{j=1}^{nK} (R^j \leq R^{j*}) \right] \geq 1 - nK(1 - \delta). \quad (3)$$

D14:

$$\Pr \left[\bigwedge_{j=1}^{nK} (|e_{j+n}^\top \sigma_{s_0} - \mathbf{F}^j(s_0)| \leq R^{j*}) \right] \geq \Delta.$$

D15:

$$\Pr \left[\bigwedge_{j=1}^{nK} (e_{j+n}^\top \sigma_{s_0} \in C_j(s_0)) \right] \geq \Delta$$

D16:

$$\Pr [\sigma_{s_0} \in \text{Zonotope}(\mathcal{F}(s_0), \mathbf{diag}([0_{1 \times n}, R^*]))] \geq \Delta \quad (4)$$

D17:

$$\text{Zonotope}(\mathcal{F}(s_0), \mathbf{diag}([0_{1 \times n}, R^*])) \subset \bar{X} \oplus \text{Zonotope}(0, \mathbf{diag}([0_{1 \times n}, R^*])) = X \quad (5)$$

D18:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ -2x_3 - 4 - \mu x_2^2 \\ x_5 \\ x_6 \\ -2x_6 + 2u - \mu x_4^2 \end{bmatrix} + v, \quad \mathcal{I} = \left\{ s_0 \left| \begin{bmatrix} 90 \\ 32 \\ 0 \\ 10 \\ 30 \\ 0 \end{bmatrix} \leq s_0 \leq \begin{bmatrix} 110 \\ 32.2 \\ 0 \\ 11 \\ 30.2 \\ 0 \end{bmatrix} \right. \right\} \quad (6)$$

D19:

$$\left\{ \begin{array}{l} \dot{x}_1 = \cos(x_8) \cos(x_9) x_4 + (\sin(x_7) \sin(x_8) \cos(x_9) - \cos(x_7) \sin(x_9)) x_5 \\ \quad + (\cos(x_7) \sin(x_8) \cos(x_9) + \sin(x_7) \sin(x_9)) x_6 + v_1 \\ \dot{x}_2 = \cos(x_8) \sin(x_9) x_4 + (\sin(x_7) \sin(x_8) \sin(x_9) + \cos(x_7) \cos(x_9)) x_5 \\ \quad + (\cos(x_7) \sin(x_8) \sin(x_9) - \sin(x_7) \cos(x_9)) x_6 + v_2 \\ \dot{x}_3 = \sin(x_8) x_4 - \sin(x_7) \cos(x_8) x_5 - \cos(x_7) \cos(x_8) x_6 + v_3 \\ \dot{x}_4 = x_{12} x_5 - x_{11} x_6 - 9.81 \sin(x_8) + v_4 \\ \dot{x}_5 = x_{10} x_6 - x_{12} x_4 + 9.81 \cos(x_8) \sin(x_7) + v_5 \\ \dot{x}_6 = x_{11} x_4 - x_{10} x_5 + 9.81 \cos(x_8) \cos(x_7) - 9.81 - u_1/1.4 + v_6 \\ \dot{x}_7 = x_{10} + (\sin(x_7)(\sin(x_8)/\cos(x_8))) x_{11} + (\cos(x_7)(\sin(x_8)/\cos(x_8))) x_{12} + v_7 \\ \dot{x}_8 = \cos(x_7) x_{11} - \sin(x_7) x_{12} + v_8 \\ \dot{x}_9 = (\sin(x_7)/\cos(x_8)) x_{11} + (\cos(x_7)/\cos(x_8)) x_{12} + v_9 \\ \dot{x}_{10} = -0.9259 x_{11} x_{12} + 18.5185 u_2 + v_{10} \\ \dot{x}_{11} = 0.9259 x_{10} x_{12} + 18.5185 u_3 + v_{11} \\ \dot{x}_{12} = v_{12} \end{array} \right. \quad \mathcal{I} = \left\{ s_0 \left| \begin{bmatrix} -0.2 \\ -0.2 \\ -0.2 \\ -0.2 \\ -0.2 \\ -0.2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \leq s_0 \leq \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right. \right\} \quad (7)$$

D20:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \\ \dot{x}_7 \end{bmatrix} = \begin{bmatrix} 1.4x_3 - 0.9x_1 \\ 2.5x_5 - 1.5x_2 \\ 0.6x_7 - 0.8x_3x_2 \\ 2.0 - 1.3x_4x_3 \\ 0.7x_1 - 1.0x_4x_5 \\ 0.3x_1 - 3.1x_6 \\ 1.8x_6 - 1.5x_7x_2 \end{bmatrix}, \quad \mathcal{I} = \left\{ s_0 \left| \begin{bmatrix} 1.05 \\ 0.9 \\ 1.35 \\ 2.25 \\ 0.85 \\ -0.05 \\ 0.3 \end{bmatrix} \leq s_0 \leq \begin{bmatrix} 1.35 \\ 1.2 \\ 1.65 \\ 2.55 \\ 1.15 \\ 0.25 \\ 0.6 \end{bmatrix} \right. \right\} \quad (8)$$

Inline

- I1: ¹
- I2: K
- I3: $\{1, 2, \dots, n\}$
- I4: $[n]$
- I5: ℓ
- I6: $[n_0, n_1, \dots, n_{\ell+1}]$
- I7: n_0
- I8: $n_{\ell+1}$
- I9: $n_i, i \in [\ell]$
- I10: i
- I11: $e_i \in \mathbb{R}^n$
- I12: $\mathbb{R}^n$
- I13: $x \sim \mathcal{X}$
- I14: x
- I15: $\mathcal{X}$
- I16: A
- I17: $|A|$
- I18: $\text{Zonotope}(b, A)$
- I19: b
- I20: $\oplus$
- I21: $\lceil x \rceil$
- I22: $x \in \mathbb{R}$

I23: $\mathcal{I} \subset \mathbb{R}^n$
 I24: M
 I25: $\sigma_{s_0} = [s_0^\top, \dots, s_K^\top]^\top \subset \mathbb{R}^{(K+1)n}$
 I26: $\mathcal{S}$
 I27: $\sigma_{s_0} \sim \mathcal{S}$
 I28: σ_{s_0}
 I29: $s_k \sim T(s' \mid s = s_{k-1})$
 I30: $\mathcal{I}$
 I31: $\mathcal{W}$
 I32: $(\Pr[s_0 \notin \mathcal{I}] = 0)$
 I33: $s_0 \overset{\mathcal{W}}{\sim} \mathcal{I}$
 I34: s_0
 I35: D_{train}
 I36: D_{test}
 I37: $D_{\text{test}} = \{s_{0,i}, \sigma_{s_0,i}\}_{i=1}^L$
 I38: L
 I39: $1 - \varepsilon$
 I40: $\varepsilon \in (0, 1)$
 I41: $\sigma_{s_0} \in \mathbb{R}^{n(K+1)}$
 I42: $\mathbb{R}^{n(K+1)}$
 I43: $\mathcal{F} : \mathbb{R}^n \rightarrow \mathbb{R}^{(K+1)n}$
 I44: n
 I45: $(K+1)n$
 I46: $\bar{\sigma}_{s_0}$
 I47: $\bar{\sigma}_{s_0} = \mathcal{F}(s_0)$
 I48: μ
 I49: $z_1, z_2, \dots, z_{m+1}$
 I50: $z_i = (x_i, y_i) \in \mathbb{R}^n \times \mathbb{R}$
 I51: $i \in [m+1]$
 I52: $\alpha \in (0, 1)$
 I53: $C(x_{m+1}) = [\mu(x_{m+1}) - R^*, \mu(x_{m+1}) + R^*] \subset \mathbb{R}$
 I54: $z_1, z_2, \dots, z_m$
 I55: $R_i = |y_i - \mu(x_i)|$
 I56: z_i
 I57: $R_1, \dots, R_{m+1}$
 I58: m
 I59: $\ell = \lceil (m+1)(1-\alpha) \rceil \leq m$
 I60: R^*
 I61: $R_1, \dots, R_m$
 I62: $R^* = R_\ell$
 I63: $\Pr[R_{m+1} \leq R^*] \geq (1-\alpha)$
 I64: $\delta = 1 - \alpha$
 I65: X
 I66: $X \subset \mathbb{R}^{n(K+1)}$
 I67: $\Delta = 1 - \varepsilon$
 I68: Δ
 I69: $\Delta \in (0, 1)$
 I70: $s_0 \overset{\mathcal{W}}{\sim} \mathcal{I}$
 I71: $X \subset \mathbb{R}^{n(K+1)}$
 I72: $\mathcal{F} : \mathbb{R}^n \rightarrow \mathbb{R}^{n(K+1)}$
 I73: $\sigma_{s_0} \in \mathbb{R}^{n(K+1)}$
 I74: $\bar{\sigma}_{s_0} \in \mathbb{R}^{n(K+1)}$
 I75: σ_{s_0}
 I76: $\mathbf{F}^j(s_0)$
 I77: $(j+n)$
 I78: $\mathcal{F}(s_0)$

I79: $\bar{X} \subset \mathbb{R}^{n(K+1)}$
 I80: $\mathcal{F}(\mathcal{I})$
 I81: $s_0 \in \mathcal{I}$
 I82: $\mathcal{F}(s_0) \in \bar{X}$
 I83: $\bar{X}$
 I84: N
 I85: (s_0, σ_{s_0})
 I86: $R^j \in \mathbb{R}_{>0}$
 I87: 2
 I88: $e_j \in \mathbb{R}^{n(K+1)}$
 I89: j^{th}
 I90: i^{th}
 I91: k
 I92: j
 I93: $i \cdot k + n$
 I94: R^j
 I95: $e_{j+n}^\top \sigma_{s_0}$
 I96: $\sigma_{s_0,i}, i \in [L]$
 I97: $\mathcal{R}_{D_{\text{test}}}^j, j \in [nK]$
 I98: $(s_{0,i}, R_i^j), i \in [L]$
 I99: $R_i^j = |e_{j+n}^\top \sigma_{s_0,i} - F^j(s_{0,i})|$
 I100: $R^j, j \in [nK]$
 I101: δ
 I102: 3
 I103: R_i^j
 I104: $R_1^j, \dots, R_L^j$
 I105: $R_1^j \leq R_2^j \leq \dots \leq R_L^j$
 I106: $R^{j*} = R_\ell^j$
 I107: $\ell = \lceil (L+1)\delta \rceil$
 I108: R^{j*}
 I109: $\inf\{z \in \mathbb{R} | \Pr[R^j \leq z] \geq \delta\}$
 I110: $\delta \in (0, 1)$
 I111: $R^* = [R^{1*}, \dots, R^{nK*}]^\top$
 I112: $\mathcal{F}$
 I113: $\delta \in (0, 1)$
 I114: $\mathcal{R}_{D_{\text{test}}}^j$
 I115: $\Delta = 1 - nK(1 - \delta)$
 I116: $\Pr[R^j \leq R^{j*}] \geq \delta$
 I117: $\Delta = 1 - nK(1 - \delta)$
 I118: $C_j(s_0)$
 I119: $[F^j(s_0) - R^{j*}, F^j(s_0) + R^{j*}]$
 I120: $s_0 \in \mathcal{I}$
 I121: $\Pr[\sigma_{s_0} \in X] \geq \Delta$
 I122: $\square$
 I123: $\Delta \in (0, 1)$
 I124: $\delta = 1 - \frac{1-\Delta}{nK}$
 I125: $\lceil (L+1)\delta \rceil \leq L$
 I126: $L \geq \lceil \frac{1+\delta}{1-\delta} \rceil$
 I127: $[k\delta t, (1+k)\delta t], k \in [K]$
 I128: $n > 3$
 I129: $v \sim \mathcal{N}(0, d^2)$
 I130: $d = \mathbf{diag}([1, 0.1, 0.05, 1, 0.1, 0.05])$
 I131: $\mu = 10^{-4}$
 I132: x_1, x_2, x_3
 I133: x_4, x_5, x_6

I134: $[6, 32, 32, 32, 6]$
 I135: u
 I136: $L = 40000$
 I137: $\delta t = 0.1$ sec
 I138: $\varepsilon = 0.01$
 I139: $\varepsilon \in [0.01, 0.1]$
 I140: $\varepsilon = 0.01$
 I141: ε
 I142: $v = [v_1, v_2, \dots, v_{12}]$
 I143: $d = \mathbf{diag}([0.05 \times \vec{1}_{1 \times 6}, 0.01 \times \vec{1}_{1 \times 6}])$
 I144: $\delta t = 0.1$ sec
 I145: $[12, 20, 20, 20, 12]$
 I146: $\varepsilon = 0.01$
 I147: 273 sec
 I148: $L = 160000$
 I149: $\delta t = 0.01$
 I150: $[7, 20, 20, 20, 7]$
 I151: $\varepsilon \in [0.01, 0.1]$